Cognizant OA - IIT Roorkee

Previous Year Questions (PYQ) - Practice Paper

Instructions: Answer all questions. Coding questions require implementation in C++23. For multiple-choice questions, select one option. Answer key is intentionally not included.

Question 1

Wind Farms - Optimal Control Center Location

Problem statement: An energy company needs to place a control center to manage a distributed wind farm. Determine the optimal location for the control center that minimizes the total connection cost.

For each wind farm i:

• The farm is located at coordinates (x[i], y[i])
• The control center is at coordinates (cx, cy)
• The distance is calculated as |x[i] - cx| + |y[i] - cy|
• The connection cost is premium[i] * distance

Example:
x = [1, 3, 2, 4]
y = [1, 2, 3, 4]
premium = [1, 3, 2, 4]

The optimal location is at position [3, 3]. The total connection cost is 17, which is the minimum possible.

Function description: Complete the function `windFarms` with parameters `int premium[n]` (distance multiplier), `int x[n]` (x coordinates), and `int y[n]` (y coordinates).

Question 2

Valid Encryption Keys

Problem statement: A cyber security firm has discovered a new type of encryption being used by a group of hackers. The encryption key will be a *valid key*, which is a number that has exactly 3 factors (or divisors). For example, 4 is valid because it has exactly 3 factors: 1, 2, and 4. But 6 is not valid because it has 4 factors: 1, 2, 3, and 6.

Given an array `keys` of length n, find the number of valid keys in the range `[l, keys[i]]`, both inclusive, for each $0 \leq i < n$.

Note:
• a is called a divisor of b if there is an integer c such that a * c = b.
• Only positive integers are taken into account for counting divisors.

Example: Given n = 2, keys = [5, 11], the answer is [1, 2].

Function description: Complete `getValidKeyCount`. It accepts `long_int keys[n]`, an array of integers, and returns `int[n]`, an array containing the answer for each query.

Constraints: $1 \leq n \leq 10^5$; $1 \leq keys[i] \leq 2.5 * 10^{13}$.

Question 3

Maximum Profit with Bitwise OR Constraint

Problem statement: Given two arrays of equal length n: `indicators` and `profit`, find the maximum possible sum of profits when selecting a subset of indicators such that their bitwise OR is less than or equal to a given integer k.

Each element in the indicators array represents a trading indicator, and the corresponding element in the profit array represents the estimated profit for that indicator.

Example:
n = 3
k = 6
indicators = [3, 4, 2]
profit = [3, 4, 5]

The optimal selection is indicators 4 and 2:
• Bitwise OR: 4 OR 2 = 6 (≤ k)
• Total profit: 4 + 5 = 9
This gives the maximum possible profit while satisfying the constraint.

Function description: Complete `getMaxProfit` with `int indicators[n]`, `int profit[n]`, and `int k`. Return the maximum possible sum of profits.

Constraints: $1 \leq n \leq 10^5$; $0 \leq \text{indicators}[i] < 2^{30}$; $1 \leq \text{profit}[i] \leq 10^9$; $0 \leq k < 2^{30}$.

Question 4

What is the main role of the encoder in a transformer?

A. To convert input tokens into rich contextual representations

B. To generate the final output text one token at a time from left to right

C. To store the model's training data

D. To split text into tokens

Question 5

Which Python data type stores key-value pairs, similar to a JSON object?

A. list, an ordered collection of items that are found by their position

B. tuple

C. dict

D. set

Question 6

What is the purpose of benchmarking an agent?

A. To measure how well it performs tasks using a fixed set of test tasks

B. To increase the number of parameters inside the underlying model

C. To translate its prompt

D. To reduce its context window

Question 7

What is bias in an AI system?

A. Systematic unfair differences in outcomes for certain groups

B. A random error that appears only once and then disappears

C. A slow response

D. The model's learning rate

Question 8

What does tool selection accuracy measure?

A. How fast a tool returns a result after it has been called

B. The cost of the tool

C. The number of tools installed

D. How often the agent chooses the right tool for a task

Question 9

Which prompt states the requirement most clearly?

A. Make me a function that works with numbers in a useful way

B. Write a Python function is_even(n) that returns True if n is even and False otherwise

C. Write a function that does the right thing for whatever input it receives

D. Python

Question 10

Why should you not hard-code API keys in source code?

A. Keys make the code run slower every time it is executed

B. Python does not allow strings

C. Anyone with access to the code can see and misuse them

D. Keys are too long

Question 11

What is task allocation?

A. Allocating memory to each agent so that it can run properly

B. Allocating money

C. Deciding which agent should perform which task

D. Naming agents

Question 12

What is concurrent execution?

A. Multiple agents working on tasks at the same time

B. One agent working at a time while all the others wait their turn

C. Sleeping between tasks

D. Rolling back tasks

Question 13

Which service model provides virtual machines, storage and networks that you manage yourself?

A. IaaS

B. SaaS, where you simply use finished software through a browser

C. PaaS

D. FaaS

Question 14

What is a cloud region?

A. A billing plan that sets the price for each resource

B. A geographic area containing one or more data centres

C. A user group

D. A programming language

Question 15

What is the output of the following code?

```
s = {1, 2, 2, 3, 3, 3}
print(len(s))
```

A. 6, because a set keeps every value that was written inside the braces

B. 4

C. 1

D. 3

Question 16

What is the difference between a class and an object?

A. An object is a blueprint, and a class is an instance created from it

B. They are two names for exactly the same thing in Python

C. A class stores only data, and an object stores only functions

D. A class is a blueprint, and an object is an instance created from it

Question 17

What is the main purpose of a decorator in Python?

A. To modify or extend a function or class without changing its source code

B. To delete a function after it runs

C. To convert every function into a class

D. To install a Python package

Question 18

Who normally manages object memory allocation and release in Python?

A. The Python runtime

B. Only the programmer using free()

C. The operating system user

D. The text editor